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Emilio Moreno-Ledesma

PHYSICS MAJOR

EDUCATION

MAJOR IN PHYSICS

Faculty of Sciences, National Autonomous University of Mexico (UNAM) : 2022-Present

Mexico City, Mexico

ABILITIES

Languages

- English (IELTS Academic 8)
- Japanese (JLPT N2 and last level at the National School of Languages, Linguistics and Translation [ENALLT])
- Spanish (native)

Programming Languages

- PYTHON
- MATLAB
- MATHEMATICA
- ARDUINOIDE, NEXTION

Soft skills

- Collaborative problem solving in research teams at undergraduate and graduate levels
- Communication skills, task organization, public speaking
- Scientific thinking, creativity and visualization
- Diligence, initiative, adaptability, attention to detail

REFERENCES

Pedro A. Quinto Su, PhD PI

Applied Optics Laboratory, Institute of Nuclear Sciences, UNAM

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Daniel Sahagún Sánchez, PhD PI

Cold Atoms and Quantum Optics Laboratory (LAFriOC), Institute of Physics, UNAM

Mail: sahagun@fisica.unam.mx

SUMMARY

Fourth year Physics student at the Faculty of Sciences, National Autonomous University of Mexico (UNAM). Strong problem solving and communication abilities tested as an intern in the Cold Atoms and Quantum Optics Laboratory, the Applied Optics Laboratory, and the Laser Optics Workshop and as a volunteer at the Electricity Workshop. Self-taught in Simulations and Data Analysis. Determined to materialize the potential of quantum technologies to change the world.

EXPERIENCE

Cold Atoms and Quantum Optics Laboratory (LAFriOC), Institute of Physics, UNAM — **January, 2024 - April, 2025**

Daniel Sahagún Sánchez, PhD PI

Intern

Design of a cateye ECDL and two rubidium ovens, characterization of a CCD camera, and implementation of vacuum systems. Training in laser optics arrangements, spectroscopy, polarimetry and atomic physics

Applied Optics Laboratory, Institute of Nuclear Sciences, UNAM — **November, 2024 - May, 2025**

Pedro A. Quinto Su, PhD PI

Intern

Design and construction of optical tweezers arrangements and characterization of their potential stiffness and shape through video analysis and QPD interferometry

Laser Optics Workshop, Faculty of Sciences, UNAM — **March, 2024 - December, 2024**

Edna M. Hernández González, PhD PI

Intern

Design and construction of a DIY spin coater, analysis of thin films with reflectometry, measurement of resistivity for silver paint and graphene under different stresses and temperatures

LXVII National Physics Congress, Chihuahua — **October, 2024**

Participant alongside A. A. Aguilar, E. M. Hernández (supervisor), D. G. Vázquez

Presentation of a DIY spin coater and PDMS thin films project

Electricity Workshop — **February, 2024 - August, 2024**

Ricardo Méndez Fragoso, PhD PI

Volunteer

Magnetic field intensity measurements through electromotive force curves. Construction and characterization of DIY capacitors

REFERENCES

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Ricardo Méndez Fragoso, PhD PI

Electricity Workshop, Physics Department Coordinator, Faculty of Science, UNAM

Mail: rich@ciencias.unam.mx
Phone : +52-56-22-4964

ICFO-UNAM-UNIANDES Frontiers Research School: Photonics for Quantum Science and Technology — **October, 2025**

Participant

***Faculty of Science Physics Congress* Organizing Comittee** — **May, 2025 - April, 2025**

Founding member

Entrepreneurship Competition: *Quantum Antarctic Startup Weekend* — **December, 2024**

First place winner